# stats101a_hw1

Jaden Chou - 206654728

2026-04-02

```r
df <- read.csv('./rainfallLAcounty2025-26.csv')
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
str(df)
```

```
## 'data.frame':    25919 obs. of  15 variables:
##  $ STATION  : chr  "USR0000CBIP" "USR0000CBIP" "USR0000CBIP" "USR0000CBIP" ...
##  $ NAME     : chr  "BIG PINES CALIFORNIA, CA US" "BIG PINES CALIFORNIA, CA US" "BIG PINES CALIFORNIA
##  $ LATITUDE : num  34.4 34.4 34.4 34.4 34.4 ...
##  $ LONGITUDE: num  -118 -118 -118 -118 -118 ...
##  $ ELEVATION: num  2108 2108 2108 2108 2108 ...
##  $ DATE     : chr  "2025-03-01" "2025-03-02" "2025-03-03" "2025-03-04" ...
##  $ AWND     : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ PRCP     : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ TMAX     : int  58 41 40 52 43 29 35 48 51 49 ...
##  $ TMIN     : int  38 22 19 33 30 20 21 23 35 32 ...
##  $ WT01     : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ WT02     : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ WT03     : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ WT05     : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ WT08     : int  NA NA NA NA NA NA NA NA NA NA ...
```

## (1)

### (a)

```r
length(unique(df))
```

```
## [1] 15
```

### (b)

```r
names(table(df$STATION))[which.min(table(df$STATION))]
```

```
## [1] "US1CALA0039"
```

### (c)

```r
max(table(df$STATION))
```

```
## [1] 377
```

## (2)

### (a)

```r
str(df)
```

```
## 'data.frame':    25919 obs. of  15 variables:
##  $ STATION  : chr  "USR0000CBIP" "USR0000CBIP" "USR0000CBIP" "USR0000CBIP" ...
##  $ NAME     : chr  "BIG PINES CALIFORNIA, CA US" "BIG PINES CALIFORNIA, CA US" "BIG PINES CALIFORNIA
##  $ LATITUDE : num  34.4 34.4 34.4 34.4 34.4 ...
##  $ LONGITUDE: num  -118 -118 -118 -118 -118 ...
##  $ ELEVATION: num  2108 2108 2108 2108 2108 ...
##  $ DATE     : chr  "2025-03-01" "2025-03-02" "2025-03-03" "2025-03-04" ...
##  $ AWND     : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ PRCP     : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ TMAX     : int  58 41 40 52 43 29 35 48 51 49 ...
##  $ TMIN     : int  38 22 19 33 30 20 21 23 35 32 ...
##  $ WT01     : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ WT02     : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ WT03     : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ WT05     : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ WT08     : int  NA NA NA NA NA NA NA NA NA NA ...
```

```r
df |>
  slice_max(DATE, with_ties=FALSE) |>
  select(DATE)
```

```
##         DATE
## 1 2026-03-13
```

```r
df |>
  slice_min(DATE, with_ties=FALSE) |>
  select(DATE)
```

```
##         DATE
## 1 2025-03-01
```

Time period is 2025-03-01 to 2026-03-13

### (b)

```r
df |>
  arrange(desc(PRCP)) |>
  slice_head() |>
  select(NAME, DATE, PRCP)
```

```
##                              NAME       DATE PRCP
## 1 SANTA CLARITA 3.1 WSW, CA US 2025-12-25 4.75
```

SANTA CLARITA 3.1 WSW, CA US 2025-12-25 4.75

## (3)

```r
ucla_stations <- df |>
  filter(NAME == "U C L A, CA US")
```

### (a)

```r
ucla_stations |>
  filter(PRCP >= 1) |>
  summarize(n = n())
```

```
##   n
## 1 5
```

5 days

### (b)

```r
ucla_stations |>
  filter(PRCP >= 1) |>
  select(PRCP) |>
  summary()
```

```
##       PRCP
##  Min.   :1.72
##  1st Qu.:1.83
##  Median :1.86
##  Mean   :2.14
##  3rd Qu.:2.55
##  Max.   :2.74
```

### (c)

```r
ucla_stations[ucla_stations$TMAX == max(ucla_stations$TMAX), ][, c('TMAX', 'DATE')]
```

```
##     TMAX       DATE
## 243   94 2025-10-29
```

```r
ucla_stations[ucla_stations$TMIN == max(ucla_stations$TMIN), ][, c('TMIN', 'DATE')]
```

```
##     TMIN       DATE
## 175   69 2025-08-22
```

hottest high temperature: 94 on 2025-10-29 hottest low temperature: 69 on 2025-08-22

## (4)

```r
prcp_df <- df |>
  group_by(NAME) |>
  summarize(avg_PRCP = mean(PRCP, na.rm = TRUE), total_PRCP = sum(PRCP, na.rm=TRUE))

prcp_df |>
  filter(avg_PRCP == max(avg_PRCP, na.rm = TRUE)) |>
  select(NAME, avg_PRCP)
```

```
## # A tibble: 1 x 2
##   NAME                         avg_PRCP
##   <chr>                           <dbl>
## 1 SANTA CLARITA 3.1 WSW, CA US    0.652
```

```r
prcp_df |>
  arrange(desc(total_PRCP)) |>
  slice_head() |>
  select(NAME, total_PRCP)
```

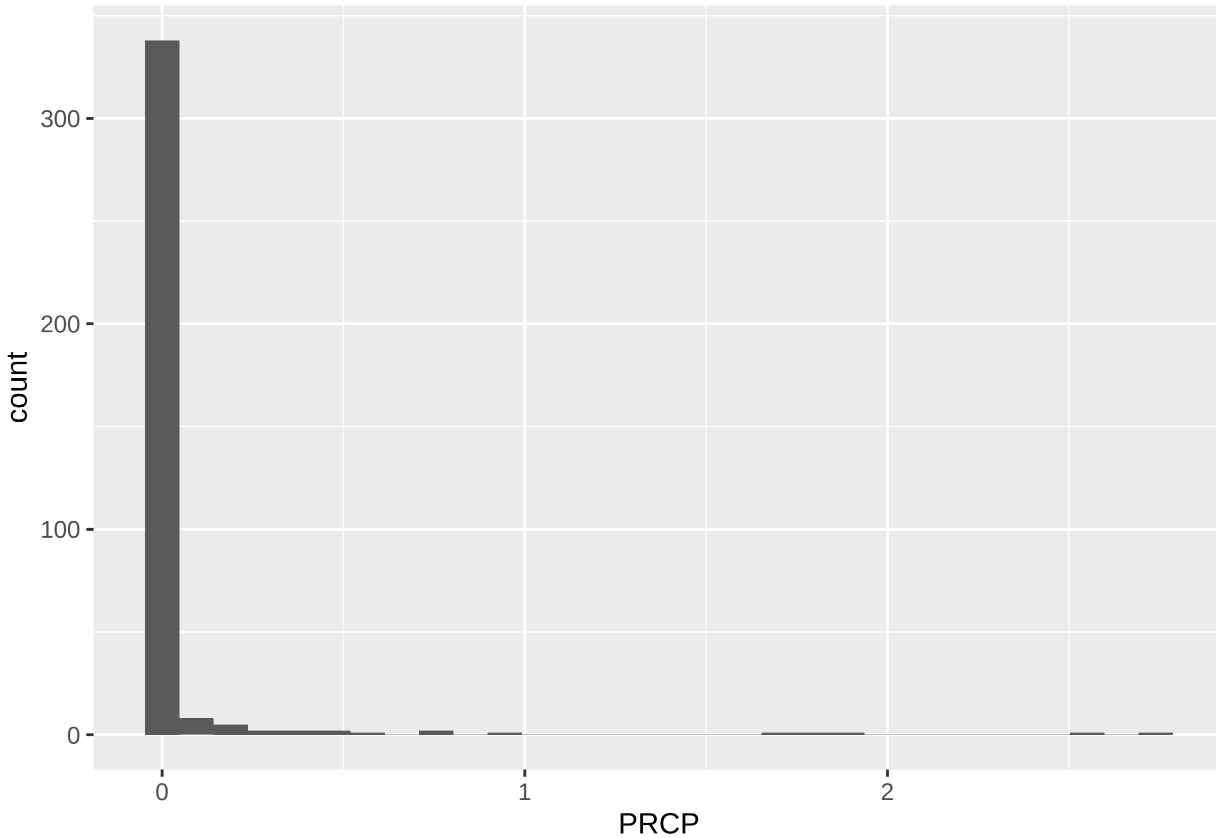
```
## # A tibble: 1 x 2
##   NAME                                 total_PRCP
##   <chr>                                     <dbl>
## 1 LA CRESCENTA MONTROSE 0.3 NE, CA US       37.9
```

highest average precipitation is SANTA CLARITA 3.1 WSW, CA US with 0.6525 inches highest total is LA CRESCENTA MONTROSE 0.3 NE, CA US with 37.86 inches

## (5)

```r
ggplot(ucla_stations, aes(x= PRCP)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




right-skewed distribution. (most values are near 0)

## (6)

### (a)

independence - samples are independent of each other. observations within each sample are independent.

random sampling - it should be random

normal distribution - both population distributions are normal or sample sizes are large.

### (b)

independence within groups - can not verify from data alone - need to check study design and procedures. - does each person's salary influence another's? - need to assume that they are independent.

independence between groups - yes, since individuals are either female or male, so no overlap. - can verify it holds

normal distribution - if the sample size is large (>= 30), Central Limit Theorem states the sampling distribution of the mean is normal. - so the sampling distribution of the difference in sample means is approx normal.

## (7)

### (a)

The syllabus states that the two lowest participation is dropped. which means we are allowed to miss two participation points and still get full credit. Which means the the percent of the total score that our participation is worth is still 1% of the total score.

### (b)

You may not turn in late homework

### (c)

No the lowest exam score is not dropped. I wish :(